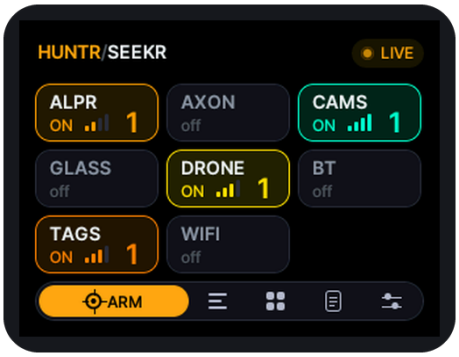
HUNTR/SEEKR

Counter-surveillance signal scanner

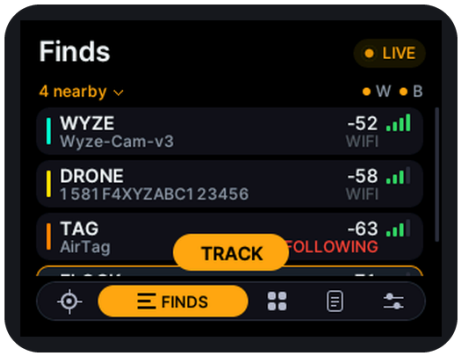
CYD 2.8 Edition
User Guide and Start-up Guide

ESP32-2432S028

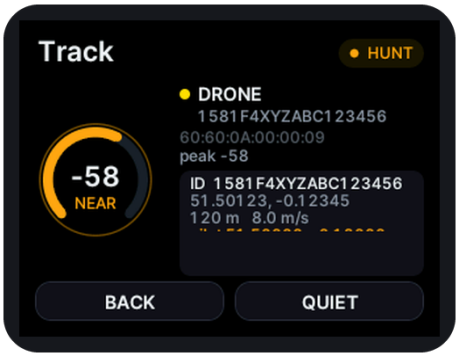
Public Beta 1



Arm groups



See what it finds



Track a signal

Printable guide. Screen pictures are example renderings of the interface; your screen will match the layout.

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1. About this device

HUNTR/SEEKR turns the ESP32-2432S028 (the 2.8 inch "cheap yellow display", CYD 2.8) board into a pocket radio scanner. It listens on the 2.4 GHz band for the wireless signals that certain kinds of devices give off, compares them with built-in lists of known makers and names, and shows you what it heard, how strong each signal is, and lets you follow a signal to its source.

What it can find

- **Wi-Fi devices** such as security cameras, doorbells, smart glasses, drones and license-plate readers, from their network names and hardware addresses. Smart bulbs in set-up mode are tagged LIGHT.
- **Bluetooth devices** such as trackers (AirTag, Tile, SmartTag, Google Find Hub), body cameras, smart glasses and drones that announce themselves.
- **Bluetooth types**: beacons (iBeacon, Eddystone, AltBeacon), wearables (watches, fitness bands, rings), smart lights, Fast Pair accessories and Bluetooth Mesh nodes, each shown with a type tag.
- **Drone Remote ID**: many drones broadcast their own serial number and position, which this device can read and display.
- **Possible card skimmers** : Bluetooth LE modules still wearing a factory name such as HC-08 or linvor, the kind sometimes hidden in gas pumps and ATM card readers. See section 6.9 for what this can and cannot do.

How it works in one minute

Every Wi-Fi access point and camera regularly broadcasts a short **beacon** that says who it is: a network name (**SSID**) and a hardware address (**MAC address**). The first three bytes of a MAC address are the **OUI**, a code that identifies the manufacturer. Bluetooth devices do something similar: they **advertise** a name and manufacturer data. HUNTR/SEEKR receives these broadcasts, matches names and OUIs against its lists, and groups the matches into **groups** such as CAMS or TAGS.

It also measures **signal strength (RSSI)** in dBm. A stronger signal usually means the device is closer, which is what lets you track one down.

> **IMPORTANT Read this once**
> Results are hints, not proof. Device names and addresses can be changed or randomized, many devices are not in the lists, and a quiet device may simply be out of range. "Nothing found" never proves nothing is there. Do not treat a match as evidence about a person or place.

> **NOTE About Public Beta 1**
> This is a **public beta**: the first release for general use. It does what this guide describes, but a beta can have rough edges and some behaviour may change in later releases. Things that depend on parts you add yourself (a speaker, a GPS module, a microSD card) work only as well as your wiring and parts allow. If something does not behave as described, section 11 lists what to check, and the project page explains how to report a problem.

2. What you need

Item	Notes
ESP32-2432S028 board	The board this guide covers (sold as ESP32-2432S028, "CYD 2.8" or "cheap yellow display"). It has a 2.8 inch resistive touchscreen, a microSD card slot, an amplifier with a socket for a small speaker, and a USB port.
USB data cable	The kind your board's USB port takes (Micro-USB on most units, USB-C on some). It must be a data cable, not a charge-only cable. Some cables that came with power banks are charge-only.
A computer	Windows, macOS or Linux, to load the software onto the board.
The firmware file	**HUNTR-SEEKR_CYD_2432S028_Public_Beta_1.bin** and **SHA256SUMS.txt**, from the project's Releases page (section 3.1).
A flashing tool or the web flasher	Either the project's **web flasher** (Chrome or Edge, nothing to install) or **esptool** (free, needs Python 3). Section 3.2 shows both.
A phone (optional)	Used for the Live view and to set the clock (this board has no clock chip) over the board's own Wi-Fi hotspot.
A speaker (optional)	The board has a small amplifier and a two-pin speaker socket, but most units are sold without a speaker. Plug in a small **8 ohm speaker** (1 watt is plenty) to hear the chimes, beeps and alarms; the software drives the amplifier from **IO26**. Without a speaker the board is silent and nothing else changes.
A microSD card (optional)	Logs, the ignore list and your own signatures are kept on the card when one is fitted at start-up; otherwise in the board's flash. **SETUP > Save to** can force either. Use a FAT32 card (32 GB or smaller is safest) and fit it before you switch the board on.
A GPS module (optional)	Connects to the 4-pin **CN1** connector: the module's TX to **IO27**, its RX to **IO22**, plus **3V3** and **GND** (9600 baud). It adds your position to the log.

> **TIP One device**
> This guide is for the **CYD 2.8** (ESP32-2432S028). The boot screen shows the software version in small text at the bottom.

3. Quick start

3.1 Get the firmware file

Download **HUNTR-SEEKR_CYD_2432S028_Public_Beta_1.bin** and **SHA256SUMS.txt** from the project's Releases page and keep them in one folder. (The web flasher in section 3.2 has the file built in, so Option A needs no download.) To check a download is intact, compare its checksum with the line in SHA256SUMS.txt (optional but recommended):

Computer	Command
Windows (PowerShell)	**Get-FileHash .\HUNTR-SEEKR_CYD_2432S028_Public_Beta_1.bin -Algorithm SHA256**
macOS or Linux	**shasum -a 256 -c SHA256SUMS.txt**

3.2 Flash it to the board

There are two ways to put the software on the board. Both write the same single file.

Option A: the web flasher (easiest)

1. On a computer, open the project's **web flasher** page (linked from the project's GitHub page) in **Google Chrome** or **Microsoft Edge**. Phones and other browsers cannot flash a board.
2. Plug the board into the computer with a USB data cable, choose the **CYD 2.8** (ESP32-2432S028) entry, and press the flash button.
3. Choose the board's serial port in the list the browser shows, then wait for the progress bar to finish. The board restarts by itself.

Option B: esptool

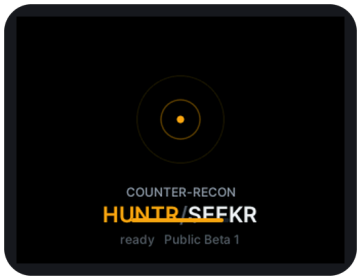
1. Install **Python 3** (python.org) if you do not have it, then install the flashing tool once: **pip install esptool** (on Windows, if pip is not found: **py -m pip install esptool**).
2. Plug the board into the computer with the USB data cable.
3. Find the port: on **Windows** it is a COM port in Device Manager (for example COM5); on **macOS** it looks like /dev/cu.usbserial-... or /dev/cu.wchusbserial... (list with ls /dev/cu.*); on **Linux** it is usually /dev/ttyUSB0.
4. In a terminal, in the folder with the file, run (replace YOUR_PORT):
   esptool --chip esp32 --port YOUR_PORT --baud 460800 write-flash 0x0 HUNTR-SEEKR_CYD_2432S028_Public_Beta_1.bin
5. Wait for **Hash of data verified** and **Hard resetting**. The board restarts by itself.

> **TIP If the computer cannot connect to the board**
> Use a **data** cable (not a charge-only one), and try another cable and another port on the computer. Most CYD boards talk to the computer through a CH340 or CP210x USB chip: if no port appears, install the free driver for that chip (WCH CH340, or Silicon Labs CP210x). Then put the chip in download mode: unplug the cable, **hold the board's BOOT button** (on the back), plug the cable in, release BOOT, and try again. If the connection is unreliable, use **--baud 115200** (Option B).

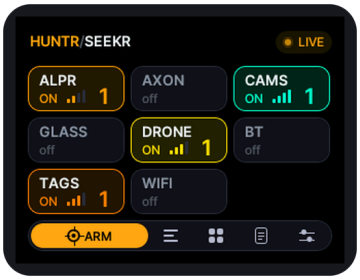
> **NOTE What flashing changes**
> Writing the file replaces the software and **resets the saved settings**, including the touch calibration, so you will be asked to press the four crosses again. Logs saved on the microSD card are not touched. To update later, flash the newer file the same way.

3.3 First start-up checklist

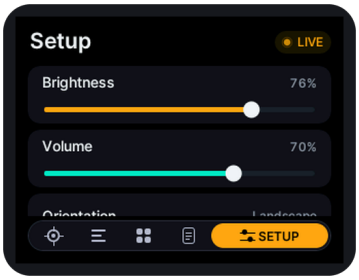
1. The screen shows a small radar animation and the name, then a line at the bottom: **ready Public Beta 1**.
2. The first time only, the **Touch setup** screen appears by itself, before anything else: press and hold each of the four crosses (section 6.8). Then the **ARM** page appears with eight group tiles. That is the home screen.
3. If the bottom line says **touch NOT detected** in red, see Troubleshooting (section 11.2).
4. Tap the tiles to check that touch works. Then go to **SETUP > Self-test** (see section 10). Every dot should be green. **Motion** is grey and says **not fitted** (this board has no motion sensor), and **Clock** is amber until you set it from your phone.



Boot screen



ARM page (home)



SETUP page

4. Getting around

The device has five pages. Swipe left and right to move between them, or tap the tab bar along the bottom. The tab you are on grows to show its name.

Page	What it is for
ARM	Turn detection groups on and off. Each group is a tile.
FINDS	The live list of everything found (strongest first by default; tap the count to change the view).
GROUPS	Every group with what it looks for, its live results, and one-tap Scenes.
LOG	The findings record (SAVE LOG), counters, the Share button and the Session Report.
SETUP	Brightness, volume, orientation, alerts and other settings.

Touch gestures

Do this	To get this
Tap	Select or press a button, arm or disarm a tile, choose a row.
Swipe left / right	Move to the next / previous page.
Drag up / down	Scroll a long list or page.
Long-press a tile (about half a second)	Open the list of finds for just that group.
Long-press a find in a list (about one second)	Hide that device. Bring it back from the Ignored screen (tap N hidden).
Tap the status pill (top right)	Pause or resume scanning.
Long-press the status pill	Switch Discreet mode on or off.
Touch a sleeping screen	The first touch only wakes the screen. Touch again to act.

The status pill

The rounded label at the top right tells you what the device is doing. (This board has no battery, so it shows no battery icon.)

Pill says	Meaning
LIVE	Scanning is running normally.
PAUSED	You paused scanning. Nothing new is recorded.
IDLE	No group is armed, so nothing is being scanned.
HUNT	You are tracking one device.
QUIET	Discreet mode is on: silent, dim, no banners.
SAVER	Power saver is active: dimmer, shorter screen time, slower scanning.
SKIM?	A possible card skimmer was heard in the last 3 minutes (section 6.9). The pill is red.

5. The detection groups

Everything the device looks for is organized into eight groups. Arm the ones you care about; the device only records matches for armed groups. (This board's chip has no Zigbee/Thread radio, so there is no ZB/TH group.)

Group	Looks for	Radio
ALPR	Flock (cameras, Raven sensors, battery packs), Penguin, Pigvision, Ubicquia and Genetec AutoVu (license-plate readers)	Wi-Fi and Bluetooth
AXON	Body cameras, docks, TASER (Axon equipment), also recognised by their Bluetooth service data and company ID	Wi-Fi and Bluetooth
CAMS	Wyze, Ring, Tapo, Reolink, Eufy, Nest, Arlo, Blink cameras; police body and dash cameras (Digital Ally, WatchGuard, i-PRO, Utility BodyWorn, Motorola VideoManager); Verkada, Avigilon	Wi-Fi and Bluetooth
GLASS	Ray-Ban Meta (Wayfarer, Oakley Meta), Spectacles, Vuzix smart glasses	Wi-Fi and Bluetooth
DRONE	DJI, Autel, Skydio, Parrot, and Remote ID broadcasts	Wi-Fi and Bluetooth
BT	Bluetooth devices by name or maker, with type tags (see section 5.1): IBEACON, EDDYSTONE, ALTBEACON, WEARABLE, LIGHT, FASTPAIR, MESH; also named when recognised: public-safety radios and vehicle light-bar controllers (Motorola, L3Harris, Tait, Feniex, SoundOff, Whelen)	Bluetooth
TAGS	Tile, AirTag, SmartTag, Google Find Hub trackers, Chipolo, Pebblebee	Bluetooth
WIFI	Any nearby 2.4 GHz Wi-Fi network (access point). Smart bulbs in set-up mode (WiZ, LIFX, Yeelight) are tagged LIGHT. Vehicle routers (Cradlepoint, Sierra Wireless, Peplink), rugged laptops and handhelds (Panasonic, Zebra, Datalogic) are named when recognised	Wi-Fi

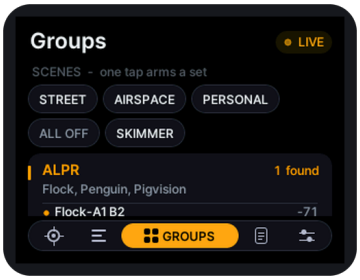
> **NOTE About the public-safety signatures**
> Many of these signatures come from a published list and are marked there as **possible**: the maker also sells ordinary products, so a match is a hint, never proof. Those are listed quietly (low confidence, no alert). What it cannot see: radios and accessories that use only **Bluetooth Classic** (many Motorola, Harris, Tait and Kenwood models), anything on 5 GHz Wi-Fi, and devices that use a random address. Agency fleet Wi-Fi names are usually changed from the maker's default, so a router can go unseen. The device recognises 166 of the 184 signatures in the list it was built from by itself (no rules file needed); the other 18 need abilities these chips lack, such as Bluetooth Classic.

Wi-Fi and Bluetooth together

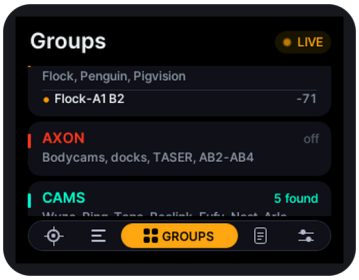
Wi-Fi and Bluetooth groups can all be armed at the same time. They share one radio and one antenna, so each may hear slightly less than it would with the radio to itself; if you only care about one kind of signal, arm just those groups. While the phone hotspot (Share) is on, Wi-Fi scanning pauses.



ARM page



GROUPS page with Scenes



A group card lists its strongest targets

Signal strength (dBm) in plain terms

Signal strength is shown as a negative number in dBm. **Closer to zero is stronger.** Each 10 dB is roughly a tenfold change in power.

Reading	What it usually means
-30 to -50	Very close: within a few metres, often the same room.
-50 to -65	Close: same room or just through a wall.
-65 to -80	Moderate: another room, or 10 to 20 metres in the open.
-80 to -90	Weak: far away or heavily blocked.
below -90	Barely detectable.

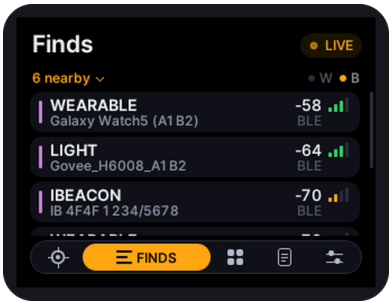
The bars beside a reading: 4 bars is stronger than -55, 3 is stronger than -65, 2 is stronger than -75, 1 is stronger than -88. Treat all of this as rough. Walls, your own body and the other device's antenna can change a reading by 10 to 20 dB.

5.1 Bluetooth type tags

Devices found by the **BT** group carry a tag in the list that says what kind of device they appear to be, instead of a plain "BT". Unnamed devices are shown by their maker (for example "Garmin" or "Huami") using the official Bluetooth company list, not a bare number.

Tag	What it means	How it is recognized
IBEACON	Apple iBeacon (shops, museums, asset tags). The row shows a short ID and the major/minor numbers, e.g. iB 4F52 1234/5678.	Apple's beacon message inside the Bluetooth advertisement.
EDDYSTONE	Google's open beacon format. Shows the beacon ID or the web address it broadcasts.	The Eddystone service and frame type (UID, URL, TLM, EID).
ALTBEACON	An open beacon format used by many vendors.	The 0xBEAC marker in the manufacturer data.
WEARABLE	Watches, fitness bands, heart-rate straps and rings (Garmin, Amazfit/Huami, Polar, Suunto, Withings, Oura, Galaxy Watch and similar).	The maker, the device name, or the device's own "I am a watch" appearance code.
LIGHT	Bluetooth bulbs and LED strips (Govee, Philips Hue, Yeelight, Nanoleaf, common strip controllers).	The maker, the name, or the Hue service. Wi-Fi bulbs in set-up mode are tagged the same way.
FASTPAIR	An Android Fast Pair accessory, typically earbuds or headphones. Not a tracker.	Google's Fast Pair service.
MESH	A Bluetooth Mesh node (often lighting or sensors).	The Mesh proxy or provisioning service.

NOTE Tags are best guesses
A device can call itself anything, and many phones and watches hide behind changing addresses and say almost nothing. An Apple Watch or iPhone usually appears only as "Apple", or not at all. Type-tagged finds are quiet: they are listed but do not trigger chimes, banners or log entries.



FINDS list with type tags: WEARABLE, LIGHT, IBEACON

6. Everyday use

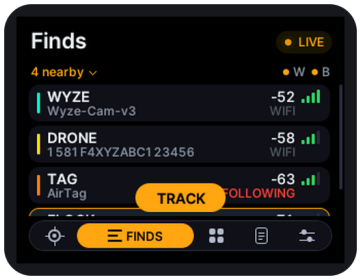
6.1 Arm groups and use Scenes

1. Go to the **ARM** page and tap a tile. It lights up and the pill changes to LIVE. Tap it again to disarm.
2. A number on the tile counts the finds in that group. The small bars on an armed tile show the strongest signal heard in that group in the last 10 seconds.
3. For a quick setup, open **GROUPS** and tap a Scene chip at the top.

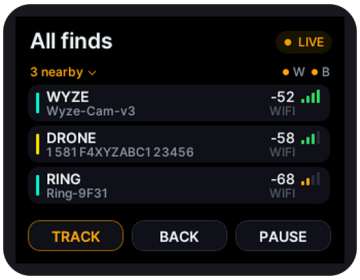
Scene	Arms
STREET	ALPR, AXON, CAMS, GLASS
AIRSPACE	DRONE
PERSONAL	TAGS and GLASS
ALL OFF	Disarms everything
SKIMMER	Bluetooth only, and switches the Skimmer alert on (section 6.9)

6.2 Read the list of finds

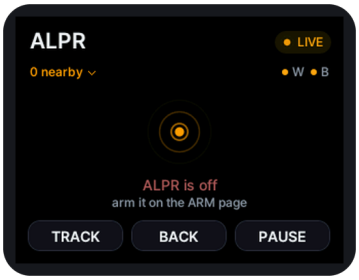
FINDS lists everything found, strongest first. Each row shows a colored group bar, the type, the device name or ID, its signal in dBm and bars, and how it was seen (WIFI or BLE). A red **FOLLOWING** tag marks a tracker that has been near you for a long time (section 6.6). Long-press a tile on the ARM page, or tap a group on the GROUPS page, to see only that group.



FINDS list



One group's list



Empty list explains why

Choosing one device in a busy area

When the list is long it is easy to tap the wrong row, so the list is built to hold still and to be narrowed:

- **It does not move while you reach for a row.** The order is frozen while a finger is on the list, for three seconds after, and while the list is scrolled down. New finds join at the end until it settles, then the list re-sorts by signal.
- **Tap the count above the list** (for example "12 nearby v") to change the view: **Strongest** (default), **Strong only** (hides anything weaker than -75 dBm and says how many it hid), **Newest** (latest find first) or **A-Z** (by name). The same choice is in **SETUP > List view** and is remembered.

- A device you have selected is never filtered out, so you can always track it. If *Strong only* hides everything, the list says so and tells you to tap the count.
- **Tap a row to select it, then tap it again** (or tap TRACK) to start tracking. Long-press to hide one (section 6.4).

6.3 Track a device

1. On **FINDS** tap the device row, then tap the amber **TRACK** button. (You can also tap a banner when a find is announced.)
2. The tracking screen shows the live signal as a big number, an arc that fills as the signal gets stronger, and a graph of the last few seconds.
3. Walk slowly. **Stronger (closer to zero) means warmer.** The word under the number changes: FAR, NEAR, then CLOSE (stronger than -55).
4. With a speaker connected it beeps faster and higher as you get closer. Tap **QUIET** to mute the beeps, **SOUND** to bring them back, and **BACK** to leave.

If the signal disappears. A Wi-Fi network is followed on its own channel, read from the network's beacon (a strong network is also heard on the channels either side of its own, so the channel the radio happened to be on is not reliable). If the network has not been heard for about two seconds, the word under the number changes to **SEARCHING** in amber, the beeps stop, and the radio sweeps outwards from its channel (+1, -1, +2, -2 and so on, across channels 1 to 13) until the network is heard again. It then stays on that channel, and moves to the beacon's stated channel if that differs. If the network really has gone, it keeps searching and picks it up as soon as it comes back. You never need to leave and restart tracking. (Bluetooth targets do not need this: they are not channel-bound in the same way.)

> **TIP Getting a good reading**
> Turn slowly on the spot. Your body blocks the signal, so the reading is strongest when the device is in front of you. Signal strength jumps around indoors, so trust the trend over a few seconds, not one number. This device cannot show a compass direction; it only tells you warmer or colder.

6.4 Hide a device you do not care about, and bring it back

Your own cameras, or a neighbour's doorbell, can fill the list. **Long-press a row** for about a second: the toast "Hidden: <name>" appears and the device is removed from the list. It will not be shown, counted, alerted on or logged again, even after a restart. Up to 32 devices can be hidden (the oldest is dropped after that).

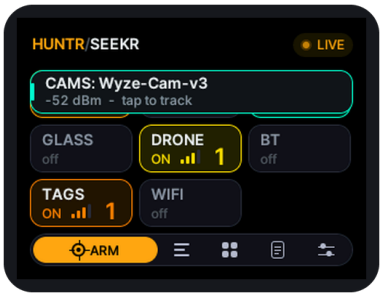
Nothing is lost. Hidden devices are kept by name on the **Ignored** screen, and there are three easy ways to reach it:

- On the **FINDS** page, tap **N hidden** next to the count at the top (shown when there is room), or the **N hidden - tap to manage** row at the end of the list.
- In **SETUP**, tap **Ignore list**.
- On the Ignored screen, tap a device to bring it back, or **RESTORE ALL**. A restored device shows up again the next time it is heard.

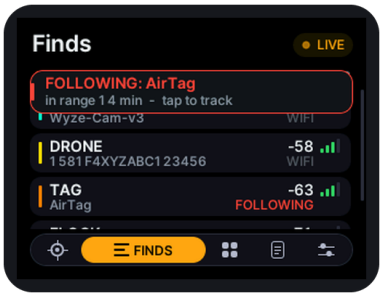
> **TIP Testing with your own network**
> If you hid your home network by accident, tap **N hidden** on the FINDS page and tap it in the list. Each hidden device is shown with its name, type and address so you can recognise it.

6.5 Alerts: banners, chimes and the find alert

- A **banner** slides in under the header for about five seconds with the group, the device name and its signal. Tap it to start tracking. A strong (high-confidence) find also wakes a sleeping screen.
- Each group has its own three-note **chime**, so you can tell what was found without looking.
- **SETUP > Find alert** chooses how often this happens: **Off**, **New group** (the default: one alert the first time each armed group finds something), or **Every find**. There is always a gap between alerts (at least four seconds, or 1.5 seconds in Every find mode), and the setting plays a sample when you change it.



Find banner



Tracker alert

6.6 Tracker alert ("following you")

A tracker that stays near you for a long time is worth noticing. When a device in the **TAGS** group has been continuously in range for the chosen time, you get a red **FOLLOWING** banner, an urgent alarm and a red tag in the list. Choose **SETUP > Tracker alert**: **5**, **10** (default) or **20** minutes, or Off. "In range" means it was heard within the last 90 seconds. It alerts once per tracker. It measures continuous time in range, so a tracker that keeps leaving and coming back may not trigger it.

6.7 Pause, quiet and power saving

Feature	How	What it does
Pause	Tap the status pill	Stops recording new finds. Tap again to resume.
Discreet mode	Long-press the pill, or SETUP > Discreet mode	Silent: no sounds, no banners, no screen wake. Screen dims after 8 seconds and turns off after 20. Brightness is capped low. Scanning and logging continue.
Power saver	SETUP > Power saver: Off / Auto / On	Screen dims after 10 seconds and turns off after 30, brightness is capped, Bluetooth scanning and Wi-Fi channel changes slow down. Auto does nothing on this board (it has no battery); choose On to use it.
Screen sleep	SETUP > Auto-dim (off by default)	When switched on: dim after 60 seconds, off after 2 minutes. A touch wakes it (the first touch only wakes). Power saver and Discreet mode dim the screen on their own, whatever this is set to.

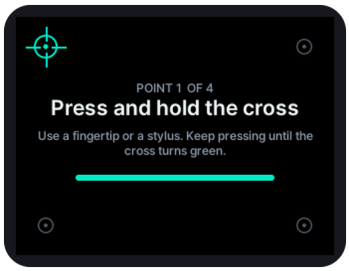
6.8 Turn the screen and calibrate touch

SETUP > Orientation cycles through **Portrait**, **Landscape** (the default, the usual way to hold a CYD), **Portrait flip** and **Landscape flip**. Landscape shows the tiles as a wide three-column grid. The choice is remembered. This board has no motion sensor, so the screen does not turn by itself.

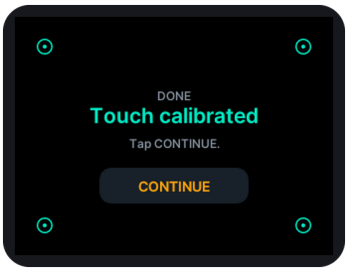
A resistive screen has to be told where its corners are. **The first time you start the board, and any time from SETUP > Calibrate touch**, the **Touch setup** screen takes over the whole display, with nothing else on it, and asks for four presses. All four targets are shown: the one to press is bright, finished ones turn green. CANCEL appears at the bottom only when you re-run it from SETUP:

1. Press and hold the cross near one corner with a fingertip or a stylus until the bar fills and the cross turns green, then lift.
2. Repeat for the other three corners. The fourth press is used to check the first three.
3. If a press was sloppy or the points do not line up, the screen says **TRY AGAIN** and starts over by itself. Your previous calibration is not overwritten by a failed attempt.
4. When it says **Touch calibrated**, tap **CONTINUE**. The result is saved in flash and works in every orientation.

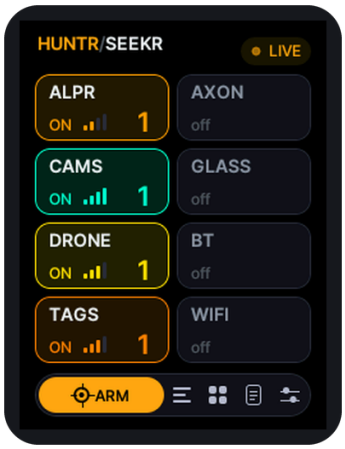
> **TIP Better touch**
> Use a stylus or a fingernail tip for the crosses and press firmly on the cross itself. If taps ever land beside the target, run Calibrate touch again.



Touch setup: press the cross



Done



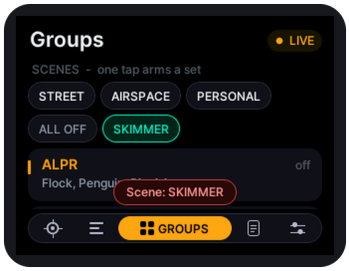
Portrait ARM page

6.9 Skimmer watch for gas pumps and ATMs

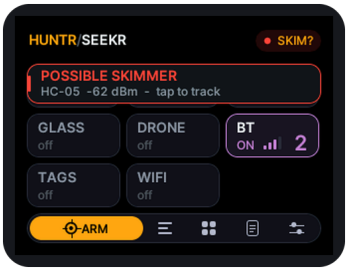
Some card skimmers are hidden inside gas pumps or ATM card readers and send the stolen card data to a thief over Bluetooth, using a cheap serial Bluetooth module that still carries its factory name. The board listens for those names and warns you loudly if it hears one.

1. Walk up to the pump or machine and open **GROUPS**, then tap the **SKIMMER** scene chip. (It arms Bluetooth and switches the Skimmer alert on. Any armed group that listens by Bluetooth works too, and the alert does not need the BT group to be armed.)
2. Stay near the pump for 10 to 20 seconds. A suspect has to be heard several times, so one stray signal does not set it off.
3. If a suspect is heard you get the warning described below. **Do not use that pump or machine.** Tap the red banner to track the signal and see which way it gets stronger.

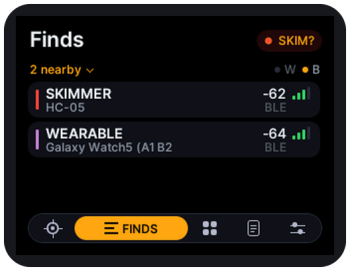
What happens	Details
Red banner	**POSSIBLE SKIMMER**, with the device name and its signal. Tap it to start tracking.
Siren	A two-tone siren from the speaker (it needs a speaker connected, section 2), about 2.5 seconds. It follows the Volume slider and is silent in Discreet mode.
Screen	A sleeping or dimmed screen wakes up.
Status pill	Turns red and says **SKIM?** for 3 minutes after the last time the device was heard.
Repeats	While the device stays in range the warning repeats about once a minute, at most four times.
Record	The device appears in FINDS with a red **SKIMMER** tag and is written into the findings record.



SKIMMER scene chip



The warning



In the FINDS list

IMPORTANT Know the limits

This only sees Bluetooth Low Energy. The most common gas-pump skimmers use *Classic* Bluetooth modules (HC-05 and HC-06), and this software listens for Bluetooth Low Energy only, so it cannot see them at all. It can catch skimmers built on a Bluetooth LE module (for example an HC-08). **A quiet screen does not mean a pump is safe.** Skimmers that store the data inside or send it by other means are invisible to this feature too.

NOTE A warning is a suspicion, not proof

It matches names only. Many harmless products and hobby projects use the same modules, and a thief can rename a module. The names watched for are HC-03, HC-05, HC-06, HC-08 (also written HC05 and HC06), linvor, and RNBT-xxxx with four hex characters; the match is exact, and capital letters do not matter.

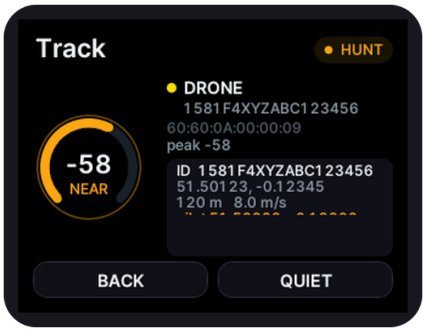
TIP If you get a warning

Do not put your card in. Move to a different pump, preferably one in the attendant's view, or pay inside. Look for signs of tampering (a loose card reader, broken security seals). Tell the attendant or the bank, and report it to your local authorities. If it keeps warning about a device you know is harmless, long-press its row in FINDS to hide it. To switch the feature off use **SETUP > Skimmer alert**.

7. Drone Remote ID

Many drones are required to broadcast **Remote ID**: a short message giving the drone's serial number, its position, height and speed, and often the pilot's position. It travels over Bluetooth or Wi-Fi and anyone can read it. HUNTR/SEEKR reads it when the **DRONE** group is armed.

- The drone's **serial number** becomes its name in the list.
- On the tracking screen a card shows the ID, the drone's latitude and longitude, its height and speed, and the **pilot's position** when it is broadcast.
- The position and pilot position are also written to the saved log and shown on the phone Live view.
- Signal strength still works as usual for finding the drone; the reported position is what the drone says about itself, and it may be missing or wrong.



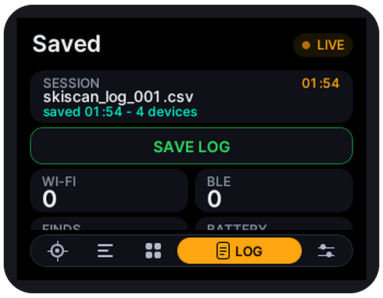
Tracking a drone with Remote ID

> **NOTE Nothing here gives you a compass arrow**
> The device has no GPS or compass, so it cannot point to a position. You can copy the coordinates into a maps app on your phone.

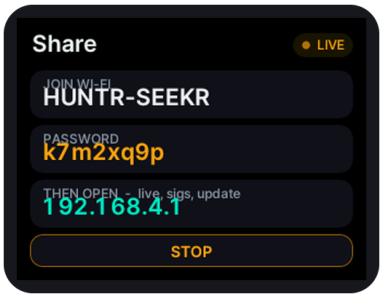
8. Sharing over Wi-Fi with your phone

The board can create its own Wi-Fi network so your phone can download logs, see a live list, edit signatures, and set the clock. It needs no internet.

1. On the board go to **LOG** and tap **SHARE**. The screen shows a network name that starts with **HUNTR-SEEKR-** and ends in four characters that change every time (for example **HUNTR-SEEKR-4F7A**), a random **password** and the address **192.168.4.1**. Both change every time you press Share, so your phone never reuses a password it saved earlier.
2. On your phone open Wi-Fi settings and join that exact network name with that password (up to two devices can join). If the phone warns there is no internet, stay connected.
3. Open a browser on the phone and go to **http://192.168.4.1**.
4. When you have finished, tap **STOP** on the board. Scanning resumes.



LOG page: SHARE and REPORT



Share screen

Web page	What you can do there
Home (192.168.4.1)	List, download and delete the saved log files ("Wipe all logs" asks first), and open the pages below.
Live view (/live)	A list of current finds that refreshes every 1.5 seconds, with signal, tracker flags and drone positions. Opening it also **sets the board's clock from your phone**.
Signatures (/sigs)	Add your own detection rules (section 9).

> **IMPORTANT While the hotspot is on**
> Wi-Fi scanning pauses because the radio is busy running the hotspot. Bluetooth scanning keeps going and everything already found stays in the list. Stop the hotspot to return to full scanning.

9. Logs, signatures, clock and updates

9.1 The findings record: save what was found, when you choose

The board keeps a **findings record** of the session in memory: every network and device it has found, one entry each, with when it was first and last heard, its best signal, channel, how many times it was heard, and where you were if a GPS is fitted. **Nothing is written to flash while you scan.** You decide when to save it, so scanning can never quietly fill the storage.

- **SAVE LOG** on the LOG page writes the record to one file, for example **skiscan_log_004.csv**. Tap it again later and the *same file is rewritten* with the new finds: it never grows by repetition.
- The LOG page's session card says exactly what is saved ("saved 21:15 - 4 devices"), or "not saved yet".
- The record holds up to 192 devices; if more are found, the ones heard longest ago make way and the toast says how many were left out. A saved file is a few tens of kilobytes at most.
- A device you have hidden (section 6.4) is not recorded. If less than 40 KB of storage is free, SAVE LOG refuses and says so instead of filling it.

SETUP > Save log chooses how saving works:

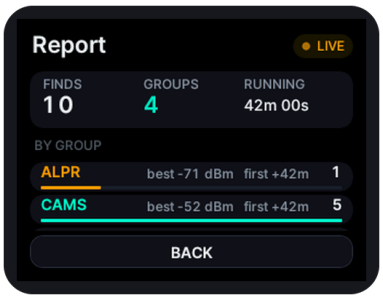
Setting	What it does
Manual (default)	Saves only when you tap SAVE LOG. Starting a new session with unsaved finds shows a reminder; nothing is written behind your back.
Auto 5 min	Refreshes the same file every 5 minutes if something changed, and whenever you stop or pause scanning. Silent unless it fails.
Live	The old way: every new find is added to a separate file (**skiscan_live_NNN.csv**) the moment it is found, once per device per session, up to 1500 rows, stopping when less than 64 KB is free. Tracking trails (**skiscan_hunt_NNN.csv**) are written only in this mode and only with a GPS fix.

Files are created only when something is saved, so starting and stopping sessions leaves no empty files behind. Get the files from the phone page (section 8), or take the microSD card out and read it on a computer. With a card fitted, logs, the ignore list and your signatures are kept on the card; without one they are kept in the board's flash. Each is a plain text table that opens in any spreadsheet. To free space, **SETUP > Storage > DELETE LOGS** (tap twice) removes the saved log files and leaves your settings, ignore list and signatures alone.

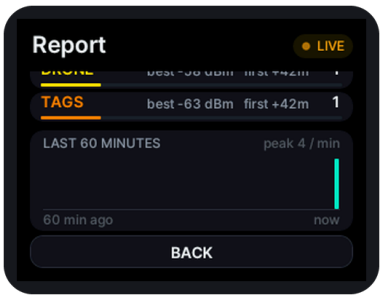
Column	Meaning
first_seen, last_seen	Date and time from the board's clock, or +HH:MM:SS from the start of the session if the clock is not set.
name	The device's network name (SSID) or ID.
mac	Its hardware address.

Column	Meaning
family, type	What it matched (for example WYZE, TAG, DRONE) and how: by hardware address (OUI), network name (SSID), Bluetooth name or type tag (WEARABLE, LIGHT...), or a Remote ID broadcast.
radio, channel	How it was heard (WIFI or BLE) and on which channel.
best_rssi, last_rssi, hits	Strongest and latest signal in dBm, and how many times it was heard.
confidence	LOW (just a nearby network or device), MED or HIGH (a strong match).
lat, lon	Where you were when it was first heard. Blank unless a GPS is fitted and has a fix.
rid_lat, rid_lon, rid_alt	The drone's own reported position and height, from Remote ID.

LOG > REPORT shows a summary of the current session on the board: total finds, the number of active groups, the running time, a bar for each group with its best signal and when it first appeared, and a chart of finds per minute over the last hour.



Session report



Finds per minute, last 60 minutes

9.2 The clock

This board has no clock chip. The clock is set from your phone: open the Live view (section 8) once and the board takes the time and time zone from the phone. It then keeps time until the board is restarted or loses power; after that it is only an estimate (from when the software was built) until you open the Live view again. The time shows at the top right of the LOG page's session card: **teal** when set from your phone, **amber** when it is only an estimate. Saved logs use it for their time stamps.

9.3 Your own detection rules (signatures)

Open **192.168.4.1/sigs** on your phone while connected to the hotspot, type rules one per line, and press Save. They are used in addition to the built-in lists and are kept after a restart. Up to 64 rules. A line that starts with # is a comment.

Rule	Example	Matches
oui	oui B4:1E:52 ALPR high	Any device whose hardware address starts with those three bytes (Wi-Fi or Bluetooth).
ssid	ssid MyCamNet CAMS	A Wi-Fi network name containing the text (not case sensitive, at least 3 letters).
ble	ble MyTag TAGS	A Bluetooth device name containing the text.
cid	cid 0x0171 GLASS	A Bluetooth manufacturer (company) ID.

After the word or code comes the **group** it should go in (ALPR, AXON, CAMS, GLASS, DRONE, BT, TAGS, WIFI, or a camera brand such as RING or WYZE) and, optionally, **low**, **med** or **high** confidence. The page shows how many rules loaded and how many lines were ignored as invalid.

9.4 Update the software

This board is updated over USB: flash the newer .bin file as described in section 3.2. The phone update page (192.168.4.1/update) is not supported on this board in this release; update over USB. Flashing resets the saved settings (including the touch calibration); saved logs stay.

9.5 A boot logo of your own

The boot screen normally shows the radar rings and the HUNTR/SEEKR title. You can replace those with your own picture. The progress bar and the version line stay.

1. Press **SHARE** on the LOG page, join the **HUNTR-SEEKR-xxxx** hotspot (the name shown on the board) and open **192.168.4.1/bootlogo** (or tap **Boot logo** on the home page).
2. Choose any picture: PNG, JPG, BMP, GIF or WebP. The page converts it in your browser: it shrinks it to fit this screen (the page states the largest size, for example 232 x 232 px on this board), turns transparent areas black, and shows you the result.
3. Press **Upload to the device**. The page says **OK - saved 232 x 116** and the boot logo is switched to **Custom** for you. **Show it on the device** displays the boot screen straight away.
4. **SETUP > Boot logo** switches between **Default** and **Custom** (only offered once a picture is stored). Each change shows a preview of the boot screen.

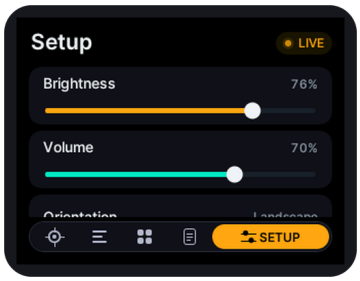
Item	Detail
Largest picture	232 x 232 pixels. The page makes the picture fit for you. A picture that is taller than the room on a given screen (for example in landscape) is shrunk to fit when it is drawn, keeping its shape; it is never enlarged.
Your existing boot.bmp	A plain **24-bit uncompressed BMP** uploaded from the hotspot's home page also works as it is, up to 1024 x 1024 (it is shrunk to fit when drawn). If both exist, the picture from the Boot logo page is used. 8-bit, compressed and other BMP types are not supported: use the Boot logo page and it will convert them.
Size on flash	A 200 x 200 picture is 78 KB. It is read from flash while the boot screen is drawn, so it uses no memory. It is drawn once; afterwards only the progress bar is redrawn.
If something goes wrong	If the file is missing, damaged or too big for the screen, the normal radar boot screen is shown instead of a blank one. Delete on the Boot logo page removes the picture and returns to Default.
Housekeeping	DELETE LOGS and Wipe all logs never touch your picture. Uploads are refused if less than 40 KB of storage would be left free.

10. Settings reference

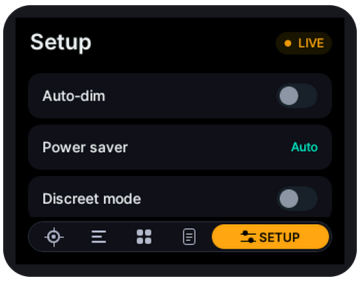
Find these on the **SETUP** page. Drag up to scroll; the list is long.

Setting	Options	Default	What it does
Brightness	Slider	About 78%	Screen brightness. The screen also dims automatically when idle.
Volume	Slider, 0 to 100%	70%	How loud the chimes and beeps are, if you have a speaker connected (section 2): all the way left is silent and it gets steadily louder towards the right. With no speaker connected it does nothing.
Orientation	Portrait, Landscape, Portrait flip, Landscape flip	Landscape	Which way the screen is drawn. The touch calibration works in every orientation.
Calibrate touch	Opens a screen	-	Runs the four-cross touch calibration again (section 6.8).
Screen colors	Normal, Swap R/B, Invert, Swap + invert	Normal	Corrects a screen whose colours look wrong. Each tap moves to the next choice and the picture changes at once; the choice is kept. **Swap R/B** fixes red and blue traded places (skin looks blue); **Invert** fixes a photo-negative picture; **Swap + invert** does both. Leave it on Normal if the colours already look right (section 11.2).
Find alert	Off, New group, Every find	New group	How often finds are announced by chime and banner.
Tracker alert	Off, 5, 10, 20 minutes	10 min	How long a tracker must stay near you before the FOLLOWING alert.

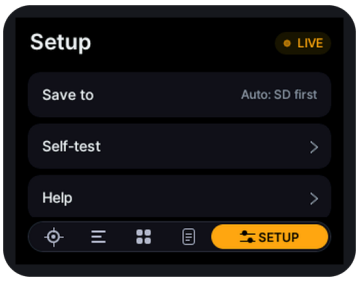
Setting	Options	Default	What it does
Skimmer alert	On / Off	On	Warns when a Bluetooth LE module with a card-skimmer name is heard (section 6.9). Works whichever groups are armed, as long as scanning is running.
Save to	Auto: SD first / SD card only / Flash only	Auto: SD first	Where logs, the ignore list and your signatures are kept. **Auto** uses the microSD card when one is fitted at start-up and the flash otherwise. **SD card only** never falls back to the flash: with no card nothing is saved, and the Self-test says so. **Flash only** ignores the card. A change applies after a restart (unplug and replug USB).
Auto-dim	On / Off	Off	When on, dims the screen after 60 s and switches it off after 2 minutes; a touch wakes it. Off by default, so the screen stays on. Turn it on to save power.
Power saver	Off, Auto, On	Auto	See section 6.7.
Discreet mode	On / Off	Off	Silent, dim, no banners. Same as a long-press on the status pill.
Boot logo	Default, Custom	Default	The picture on the boot screen. Custom is your own picture uploaded from the phone page (section 9.5).
Ignore list	Tap to open the Ignored screen	Empty	Shows how many devices are hidden. Tap a device there to restore it, or RESTORE ALL.
Self-test	Opens a screen	-	Checks every part of the hardware (next page).
Help	Opens a screen	-	A short in-device reminder of the groups and tips.
Storage	Opens a screen	-	Shows how full the flash memory is and lists the files on it.



Brightness, volume, orientation



Alerts and modes



Save to, Self-test, Help

11. Self-test and troubleshooting

11.1 The Self-test screen

SETUP > Self-test lists each part of the board with a green dot when it looks healthy and a red dot when it does not. It updates several times a second. On this board **Motion** is grey and says "not fitted" (there is no motion sensor) and **Clock** is amber until you set it from your phone. The list is longer than the screen: **drag it up and down** to see every row. **BEEP** plays the start-up melody so you can check the speaker.

Row	What green means	If it is red
Touch	The touch chip answers. Numbers show how many readings and touch signals it has sent.	Restart the board. If it stays red, flash the .bin again (section 3.2).
Display	Shows the screen size and brightness.	-
Sound	The speaker output is set up (BEEP plays the start-up melody). The board cannot tell whether a speaker is really connected.	Check the Volume slider is above zero and a speaker is plugged in (section 2); flash the .bin again.
Motion	This board has no motion sensor, so this row is grey and says "not fitted on this board". That is normal.	Nothing to fix.
Clock	The clock has a valid time.	Open the Live view from your phone to set it ("approx" means estimated).
Battery	Says "no battery detected": normal, the board runs from USB power.	Nothing to fix.
Radios	Counts of Wi-Fi and Bluetooth signals received.	Zeros with a group armed: see the Wi-Fi row below.
SD card / Storage / Memory	**SD card**: its size and free space, green when findings are saved on it. **Storage**: space used on whichever storage is in use (the card or the flash), custom rules loaded, hidden devices. **Memory**: free memory and uptime.	**SD card** red: no card was found (or Save to is Flash only), so findings go to the flash (or nowhere, with SD card only). **Storage** red means logs cannot be saved. Free space with SETUP > Storage > DELETE LOGS.

11.2 Troubleshooting

What you see	What to try
Nothing shows on the screen	Check the cable is a data cable, try another USB port and another cable, and unplug and replug the board. If it stays black or shows noise, flash the .bin again (section 3.2). If it still does, or the colours look wrong or inverted, your board may use a different display chip than this release is built for (it targets the common ILI9341 type; some units, often those with two USB sockets, use an ST7789): report the board model on the project page.
The picture appears but the colours are wrong	Open **SETUP > Screen colors** and tap it until the colours look right: **Swap R/B** if red and blue are traded, **Invert** if the picture looks like a photo negative, **Swap + invert** if it is both. Some boards use a slightly different display panel, and this makes up for it. The choice is kept.
Touch does nothing	Boot screen says "touch NOT detected"? Unplug, wait, plug in again. Check Self-test > Touch. Flash the .bin again (section 3.2).
Touch lands in the wrong place, or only works near the edges	Run **SETUP > Calibrate touch** (section 6.8) and press each cross exactly.
The first touch after a pause does nothing	Normal: that touch wakes the screen. Touch again.
No sound	Check, in this order: (1) a small speaker is plugged into the board's speaker socket (section 2); (2) the Volume slider is up and Discreet mode is off; (3) SETUP > Self-test > BEEP. The Serial Monitor (115200) prints "sound: buzzer ready on IO26" at start-up.
Wi-Fi groups find nothing	The list says "Wi-Fi radio: no frames yet" at first, then after 10 seconds "scanning access points". In that mode only access points are seen, in bursts. Wi-Fi also pauses while the Share hotspot is on. Make sure a Wi-Fi group is armed.
The list says "Wi-Fi hears nothing: unplug USB power 10 s"	The board's radio has stopped hearing Wi-Fi at all, and a restart does not always clear it: unplug USB power for about 10 seconds and plug it back in, then arm a Wi-Fi group first.
A message says "Bluetooth off: low memory"	This board has little spare memory while Wi-Fi, Bluetooth and the phone hotspot all run, so the software skips Bluetooth instead of risking a restart. Stop the Share hotspot if it is on, disarm some groups, or switch the board off and on and arm fewer groups at once. **SETUP > Self-test > Memory** shows how much memory is free.
Log times are blank or amber	The clock is not set. Open the Live view from your phone once.
My logs are not on the SD card	Open **SETUP > Self-test** and drag down to the **SD card** row: it says whether the card was found. Fit the card **before** you switch the board on (it is not noticed later). Use a **FAT32** card: cards over 32 GB are often exFAT, which this board cannot read. Check **SETUP > Save to** is not Flash only.
The phone will not stay on the hotspot	Some phones drop networks without internet. Turn off "auto-switch to mobile data" or similar while connected. The address is http://192.168.4.1 (not https).

What you see	What to try
The phone says "Unable to join the network"	Join the exact name shown on the board: it ends in four characters that change every time you press Share, along with the password. If the phone still refuses, it may have saved an older network from a previous version (or another board) that is called plain HUNTR-SEEKR: forget it (iPhone: Settings > Wi-Fi > the i next to the name > Forget This Network; Android: touch and hold the name > Forget) and join again.
The phone update page does not work	Not supported on this board in this release. Update over USB (section 3.2).
A watch or phone shows only "Apple" or does not appear	Normal. Apple devices rotate their address and send very little, so they cannot be told apart from each other.
A light or watch shows as plain BT	Its name and maker did not match the built-in lists. You can add your own rule in /sigs (section 9.3).

12. Care, safety and responsible use

- **It mostly listens.** It receives broadcasts. The access-point scan and Bluetooth scanning send the same small "who is there?" requests any phone sends. It does not jam, connect to, or interfere with other devices.
- **Know the law where you are.** Rules about scanning radio signals, recording device information and following people differ between countries and places. You are responsible for how you use it.
- **Results are not proof.** A match means a signal looked like a known kind of device. It cannot tell you whether a device is recording, who owns it, or why it is there. Names and addresses can be changed or randomized.
- **Do not use it to harass or confront anyone.** If a tracker alert worries you, move to a safe public place and contact the appropriate authorities or support services in your area.
- **Power.** Use a proper USB supply and cable. If you add your own battery or power bank, follow its maker's safety instructions.
- **Keep the software up to date** by flashing newer releases.

Quick-reference card

Cut out or fold this page and keep it with the device.

MOVE Swipe left / right, or tap the bottom tabs: ARM, FINDS, GROUPS, LOG, SETUP.	**ARM** ARM page: tap a tile. Or GROUPS page: tap a Scene chip.	**TRACK** FINDS: tap a device, then TRACK. Number closer to zero = closer to it.
HIDE A DEVICE Long-press its row for one second.	**PAUSE / RESUME** Tap the pill at the top right.	**SILENT + DIM** Long-press the pill (it shows QUIET).
WI-FI + BT Arm any mix of groups. They share one radio, so each hears a little less.	**SIGNAL (dBm)** -30 to -50 very close -50 to -65 close -65 to -80 medium -80 to -90 far	**HOTSPOT** LOG > SHARE. Join HUNTR-SEEKR-xxxx (the name shown on the board), open http://192.168.4.1. Wi-Fi scanning pauses meanwhile.
SET THE CLOCK Open /live from your phone once while on the hotspot.	**CALIBRATE TOUCH** SETUP > Calibrate touch: press and hold each of the four crosses.	**SOMETHING WRONG?** SETUP > Self-test. A red dot marks the problem.

Glossary

Term	Meaning
Access point (AP)	A device that provides a Wi-Fi network, such as a router or a camera that creates its own network.
Advertisement	A short Bluetooth message a device sends repeatedly to announce itself.
Beacon	The Wi-Fi equivalent: an access point announcing its name and details several times a second.
BLE	Bluetooth Low Energy, the power-saving kind of Bluetooth used by trackers and wearables.
Channel	A slice of the Wi-Fi band. The device steps through channels 1 to 13, listening on each briefly.
CSV	Comma-separated values: a plain text table that opens in any spreadsheet.
dBm	The unit of signal strength. Negative numbers; closer to zero is stronger.
Beacon types	iBeacon, Eddystone and AltBeacon are small, fixed Bluetooth messages that shops and venues broadcast to mark a place or object.
Fast Pair	Google's quick-pairing system for Android accessories such as earbuds.
Find Hub	Google's tracker network (formerly Find My Device). Its trackers broadcast a special Bluetooth frame that this device recognizes.
Hotspot	A Wi-Fi network created by the board itself (HUNTR-SEEKR-xxxx, a new name every time) so your phone can connect to it.
IP address (192.168.4.1)	The board's address on its own hotspot network. Type it into a phone browser.
MAC address	A device's hardware address, for example 2C:AA:8E:01:02:03.
OUI	The first three bytes of a MAC address; identify the manufacturer.
Remote ID	A broadcast from a drone giving its serial number, position and the pilot's position.
RSSI	Received signal strength indicator, the number shown in dBm.
SSID	The name of a Wi-Fi network.
Skimmer	A device hidden inside a card reader (gas pump, ATM) that copies card data. Some send it to the thief over Bluetooth.

Appendix A. Technical summary

Item	Detail
Board	ESP32-2432S028
Display	2.8 inch LCD, 320 x 240 pixels, ILI9341 driver, resistive touch (XPT2046)
Radios	2.4 GHz Wi-Fi (802.11 b/g/n) and Bluetooth 4.2 (the app uses Bluetooth LE only), one shared radio. This chip has no 5 GHz and no 802.15.4 (Zigbee / Thread) radio.
Built-in extras	microSD slot, RGB LED and light sensor (not used by the app), amplifier with a speaker socket (driven from IO26), connector CN1 for an optional GPS module. No motion sensor, no clock chip, no battery gauge.
Storage	Logs and settings on the microSD card if one is fitted, otherwise in on-board flash (about 1 MB)
Hotspot	WPA2, name HUNTR-SEEKR-xxxx (four new characters every Share), random 8-character password, address 192.168.4.1
Software	Firmware image for the ESP32. Public Beta 1